# Hardware Security

# 硬體安全

## Differential Power Analysis

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# Outline

- General description
- Attacks based on the correlation coefficient
- Calculation and simulation of correlation coefficient
- Assessing the number of power traces needed
- Alternatives to the correlation coefficient

# General Description of DPA

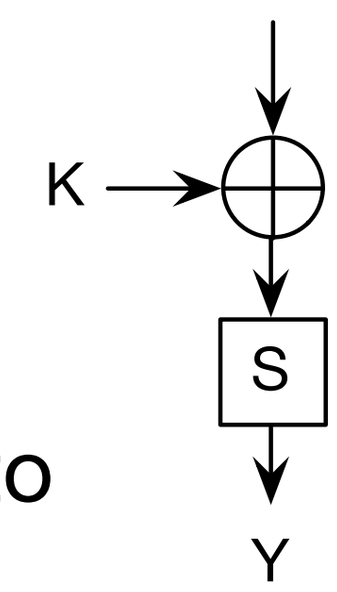
- DPA uses a large number of traces to analyze the power consumption at a fixed moment as a function of the processed data.
- Does not require detailed knowledge about the attacked device.
- Combats the noises with the strength of statistical techniques.

# DPA Example

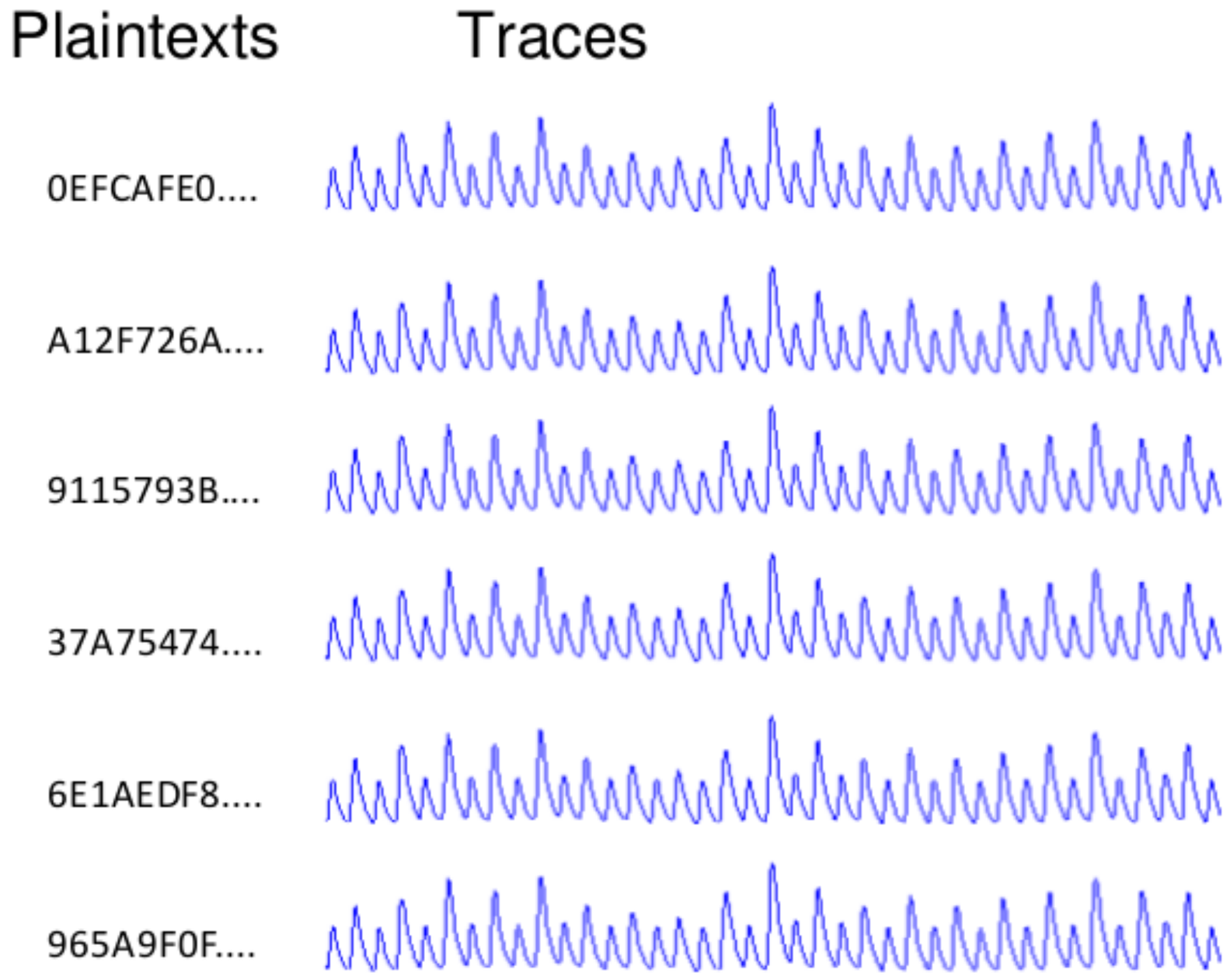
- **1-bit DPA on AES-128 encryption**
  - Until first MixColumns, each byte of state depends on one plaintext byte and one key byte
  - Recover the key byte-by-byte

x

- **For each possible value of K**
  - Compute Y for each input
  - Group traces into two subsets according to LSB(Y)
  - Compute difference of mean traces



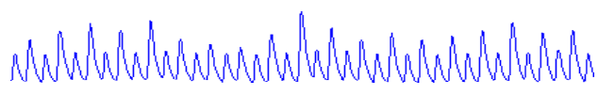
# Collecting Traces



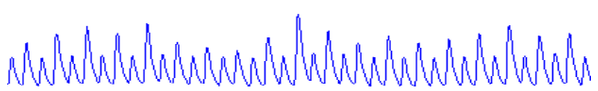
# Grouping Traces (wrong key)

If K = 0x00

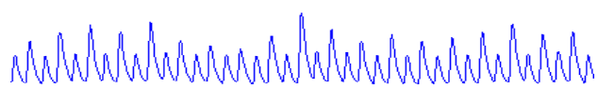
Y = 0xAB, LSB = 1



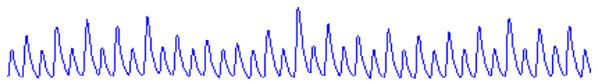
Y = 0x32, LSB = 0



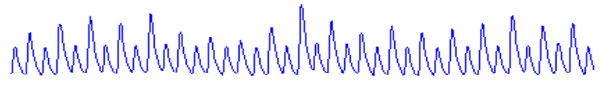
Y = 0x81, LSB = 1



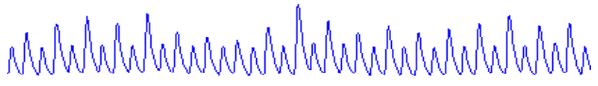
Y = 0x9A, LSB = 0



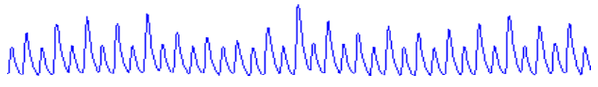
Y = 0x9F, LSB = 1



Y = 0x90, LSB = 0



Average





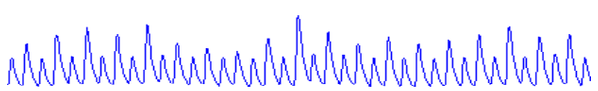
Difference



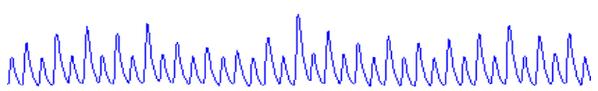
# Grouping Traces (correct key)

If K = 0x2B

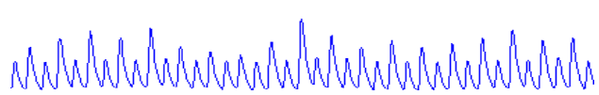
Y = 0x32, LSB = 0



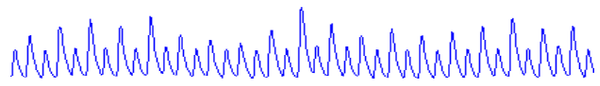
Y = 0xFC, LSB = 0



Y = 0x69, LSB = 1



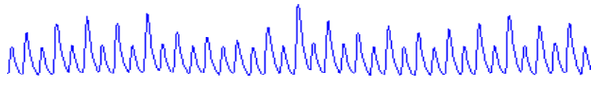
Y = 0xB7, LSB = 1



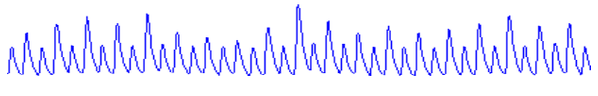
Y = 0x2D, LSB = 1

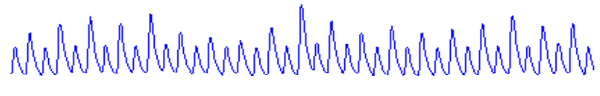


Y = 0x70, LSB = 0

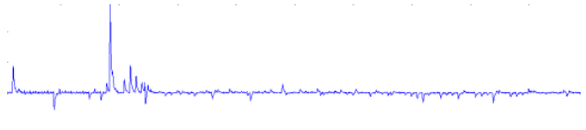


Average



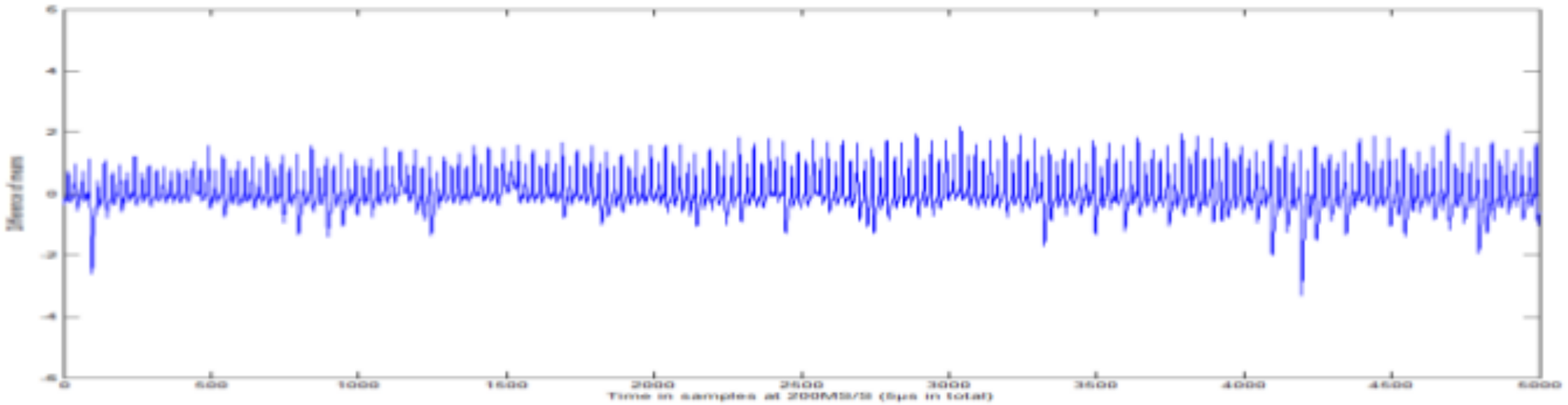


Difference

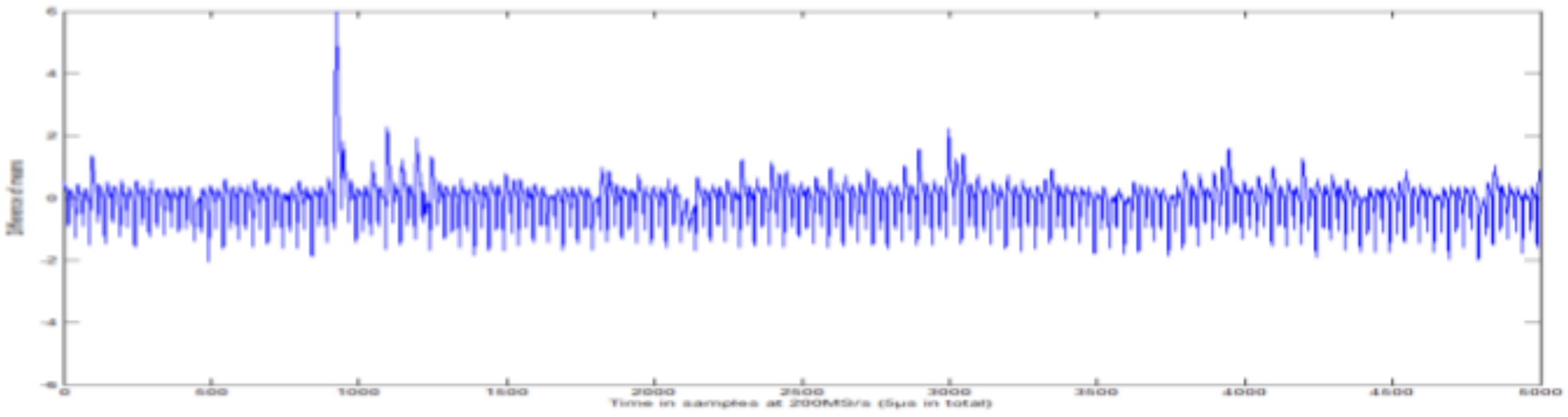


# Differential Traces

Differential trace for a wrong hypothesis on K

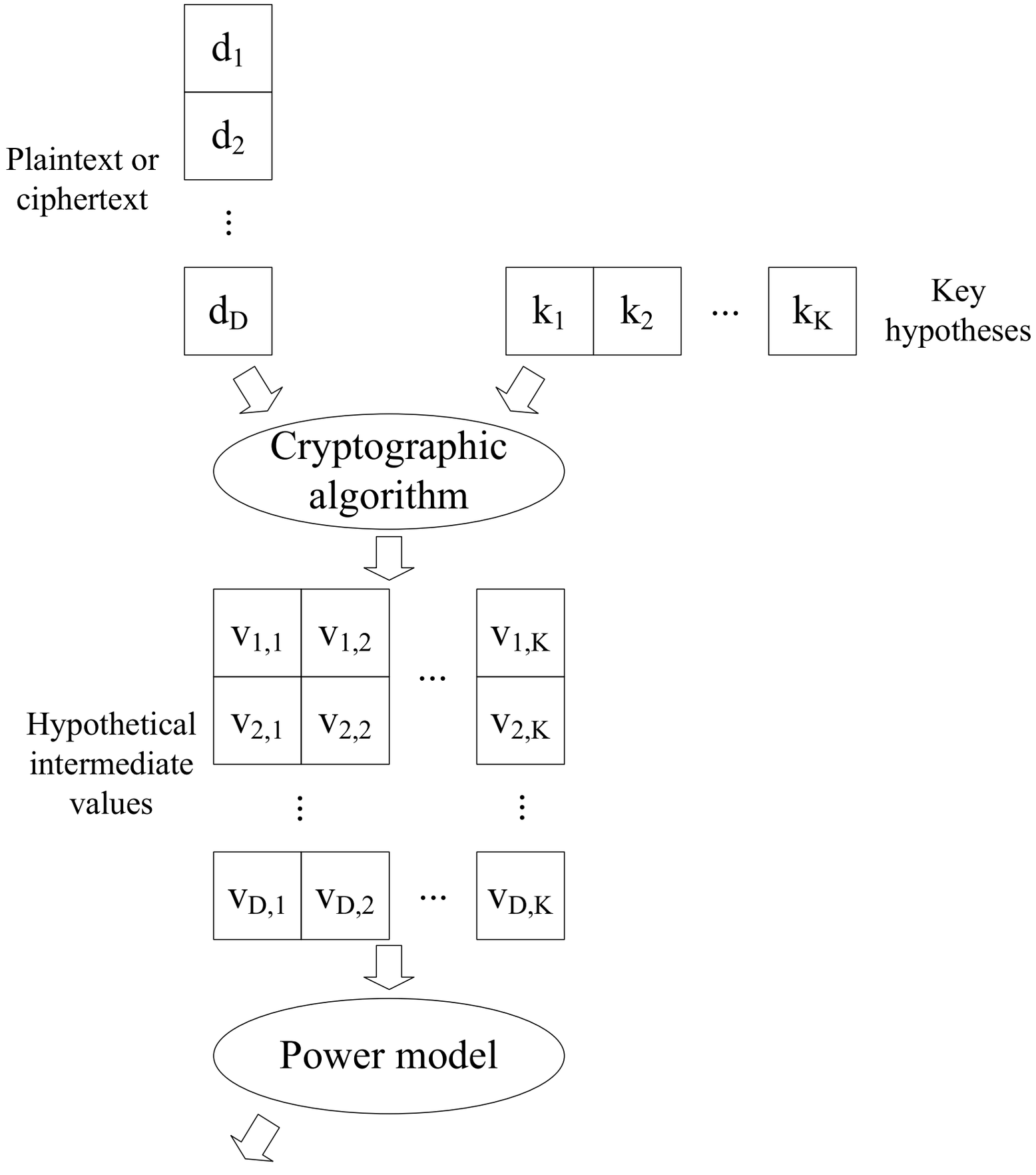


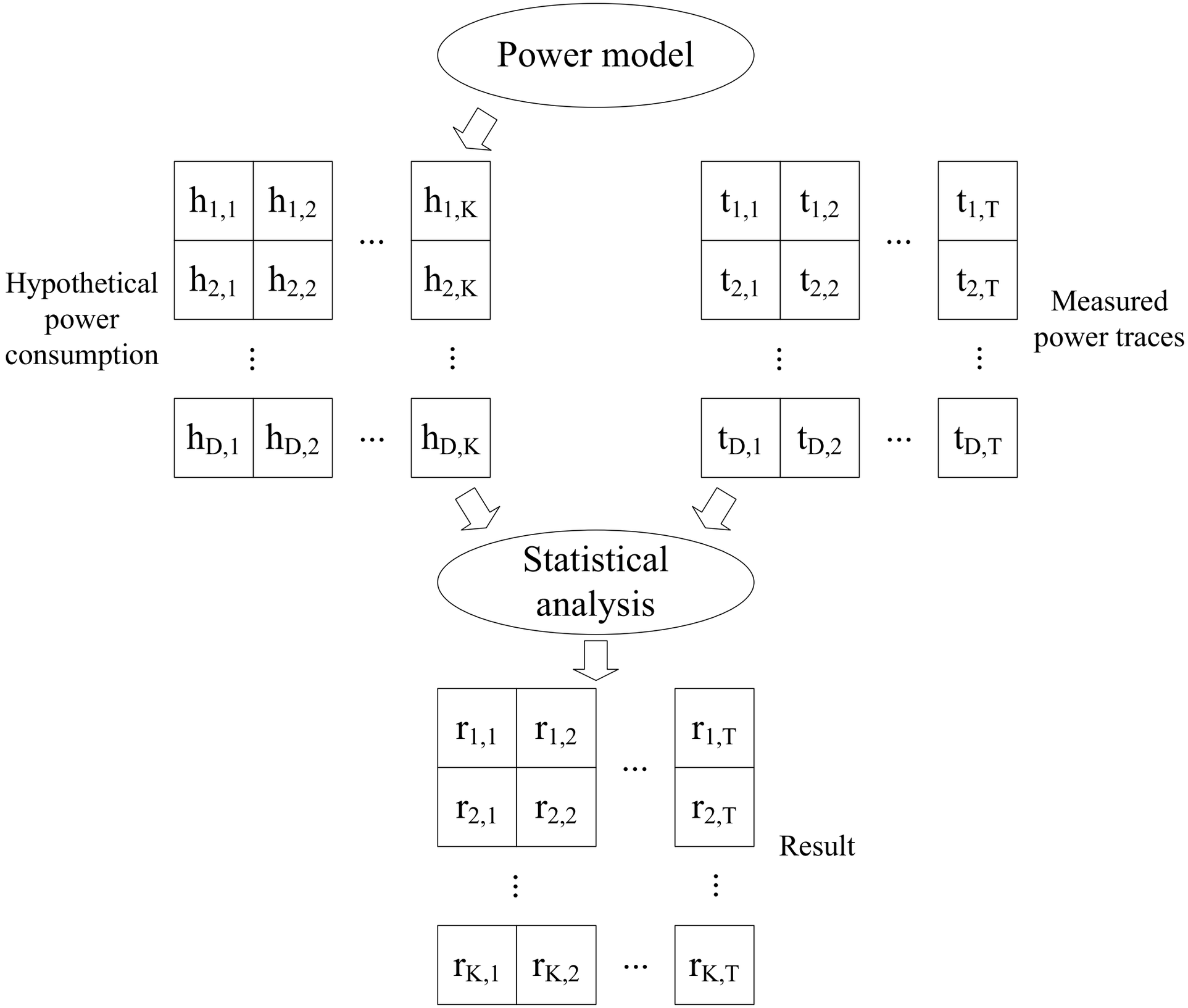

Differential trace for correct hypothesis on K



# The Attack Strategy of DPA

1. Choosing an intermediate result of the executed algorithm
2. Measuring the power consumption
3. Calculating hypothetical intermediate values
4. Mapping intermediate values to power consumption values
5. Comparing the hypothetical power consumption values with the power traces


Plaintext or ciphertext
$d_1$
$d_2$
$d_D$
$k_1$
$k_2$
$k_K$
Key hypotheses
Cryptographic algorithm
Hypothetical intermediate values
$v_{1,1}$
$v_{1,2}$
$v_{1,K}$
$v_{2,1}$
$v_{2,2}$
$v_{2,K}$
$v_{D,1}$
$v_{D,2}$
$v_{D,K}$
Power model


Power model
Hypothetical power consumption
Measured power traces
Statistical analysis
Result

# Correlation Power Analysis

- The 1-bit DPA is easy to understand, but requires very large amount of traces.
- Improved attacks based on powerful hypothetical intermediate values
- **Use of more sophisticated statistics**
  - Correlation coefficients

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

# Example of CPA on Software Implementation

- Intermediate value

$$v_{i,j} = S(d_i \oplus k_j)$$

- Measure power traces of 1000 different plaintexts
- Ways of mapping intermediate values to traces
  - LSB only
  - Hamming weight model

# Correlation of Different Key Hypotheses

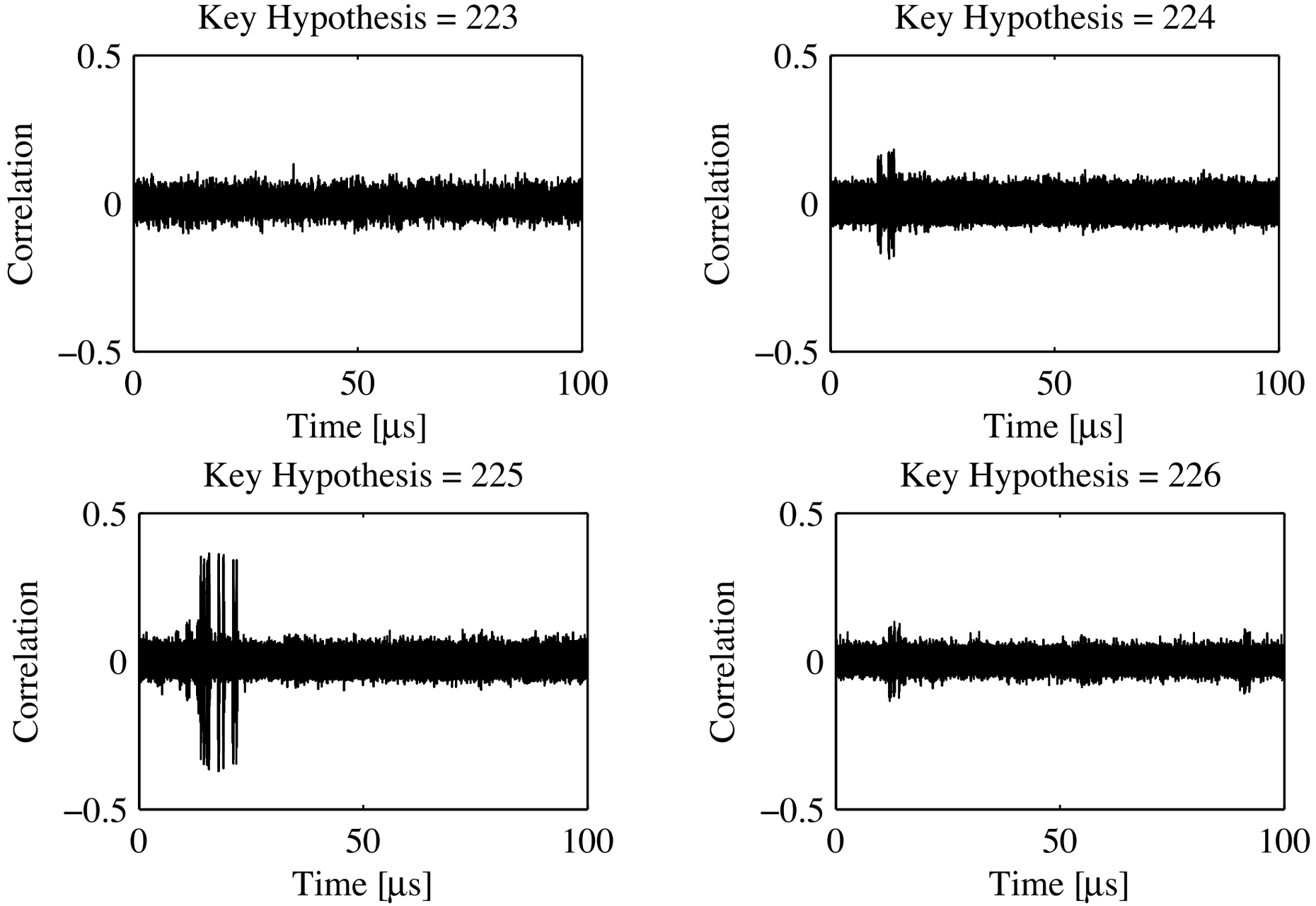
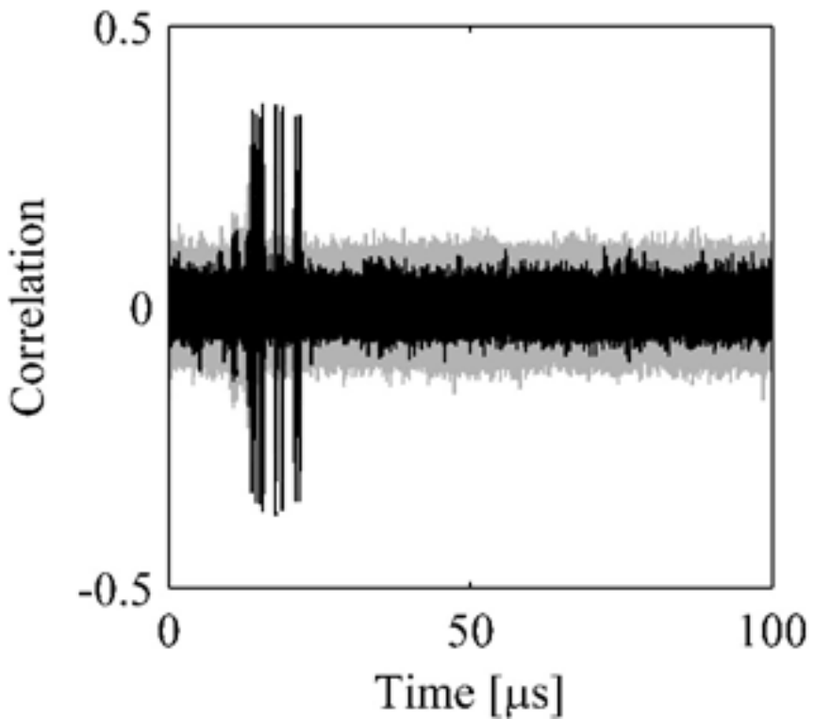
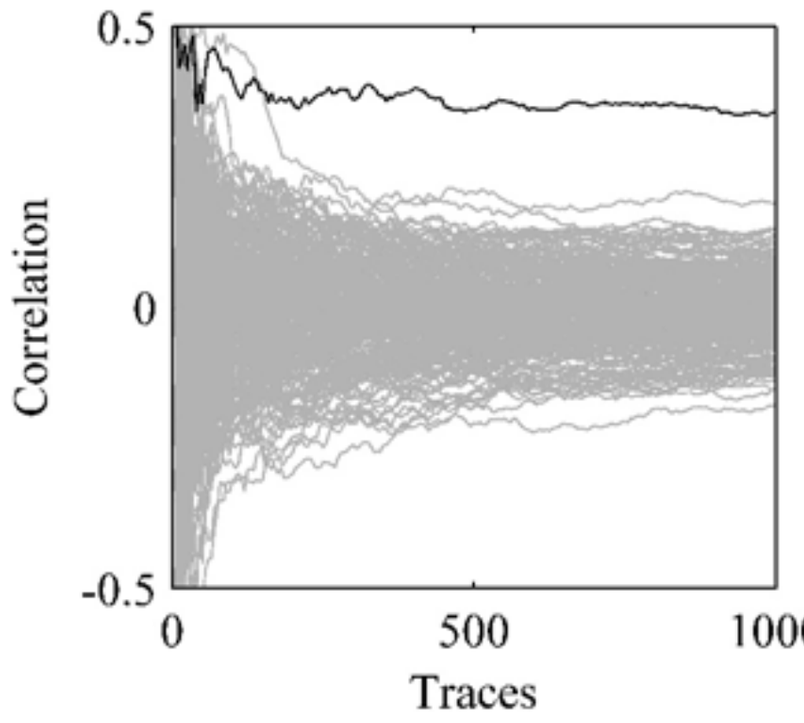


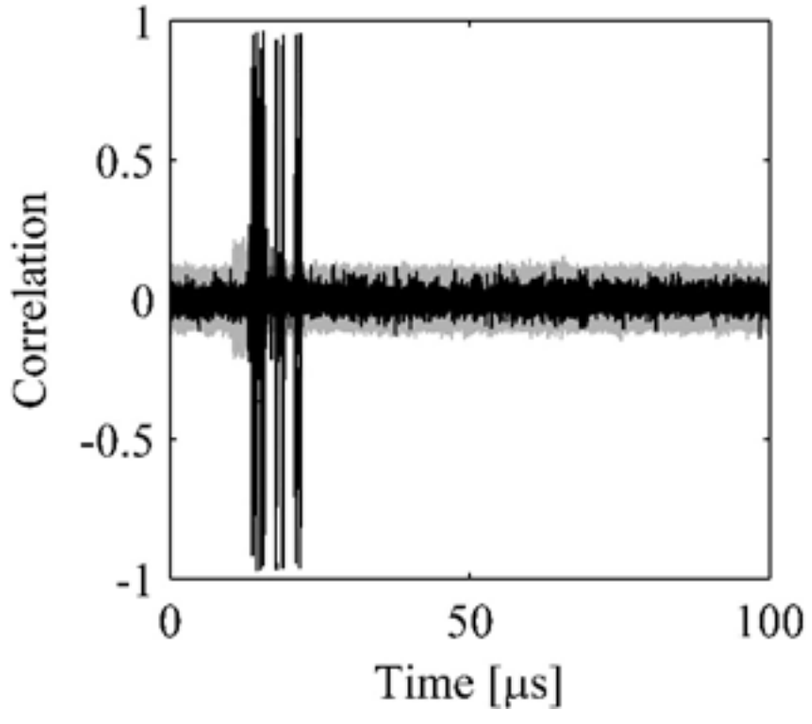

*Figure 6.2.* The rows of the matrix **R** that correspond to the key hypotheses 223, 224, 225, and 226.

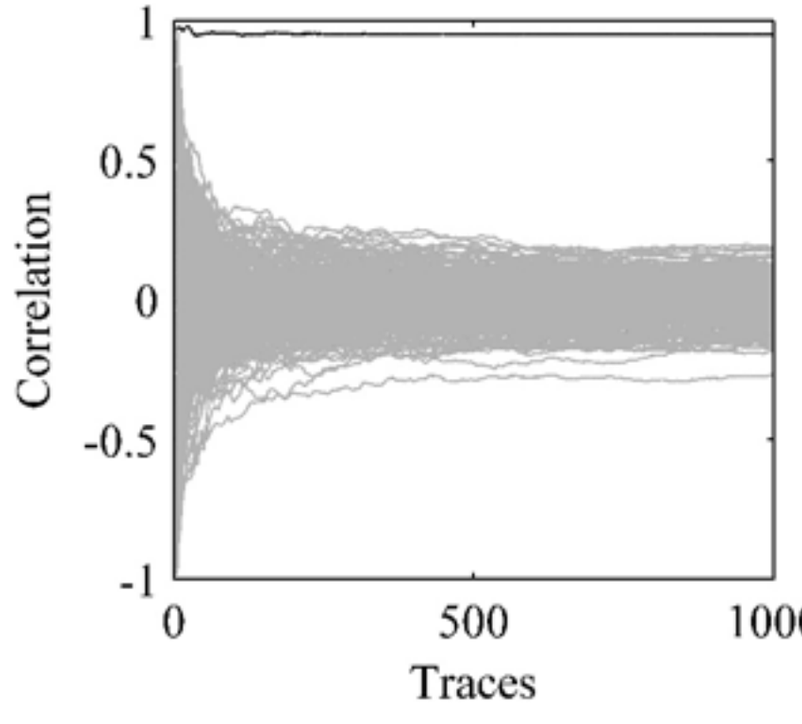
# Comparisons of Using Different Intermediate Values



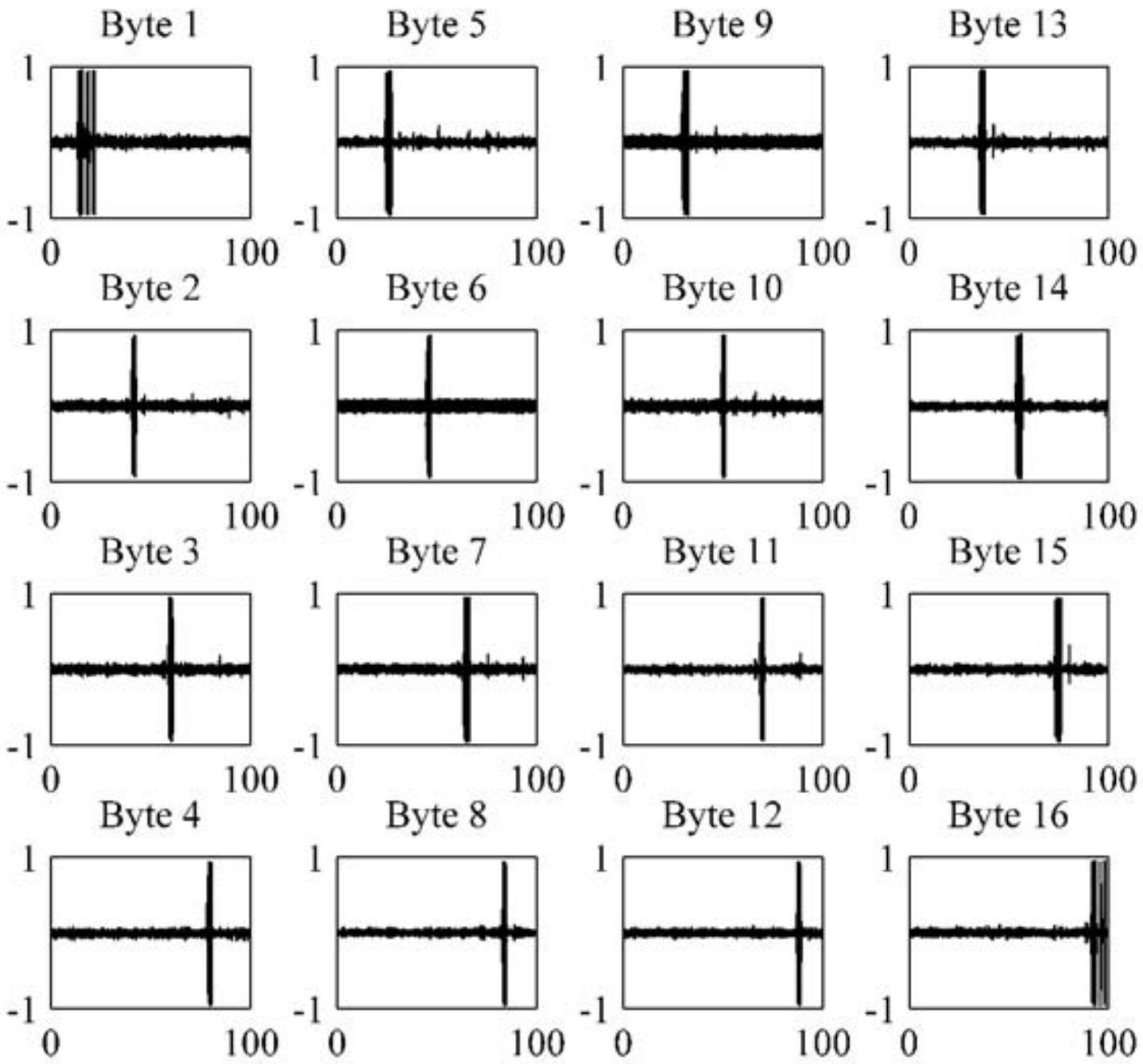

LSB only
160 traces
at least




HW model
20 traces
at least

# DPA Results on the 16 Key Bytes (Software)




- All 16 attacks have similar correlation values.
- The 16 S-box lookups are done sequentially.

*Figure 6.7.* Plots for the correct key hypotheses in the DPA attacks on the 16 S-box outputs in the first round. The y-axes show the correlation and the x-axes show the time in $\mu$s.

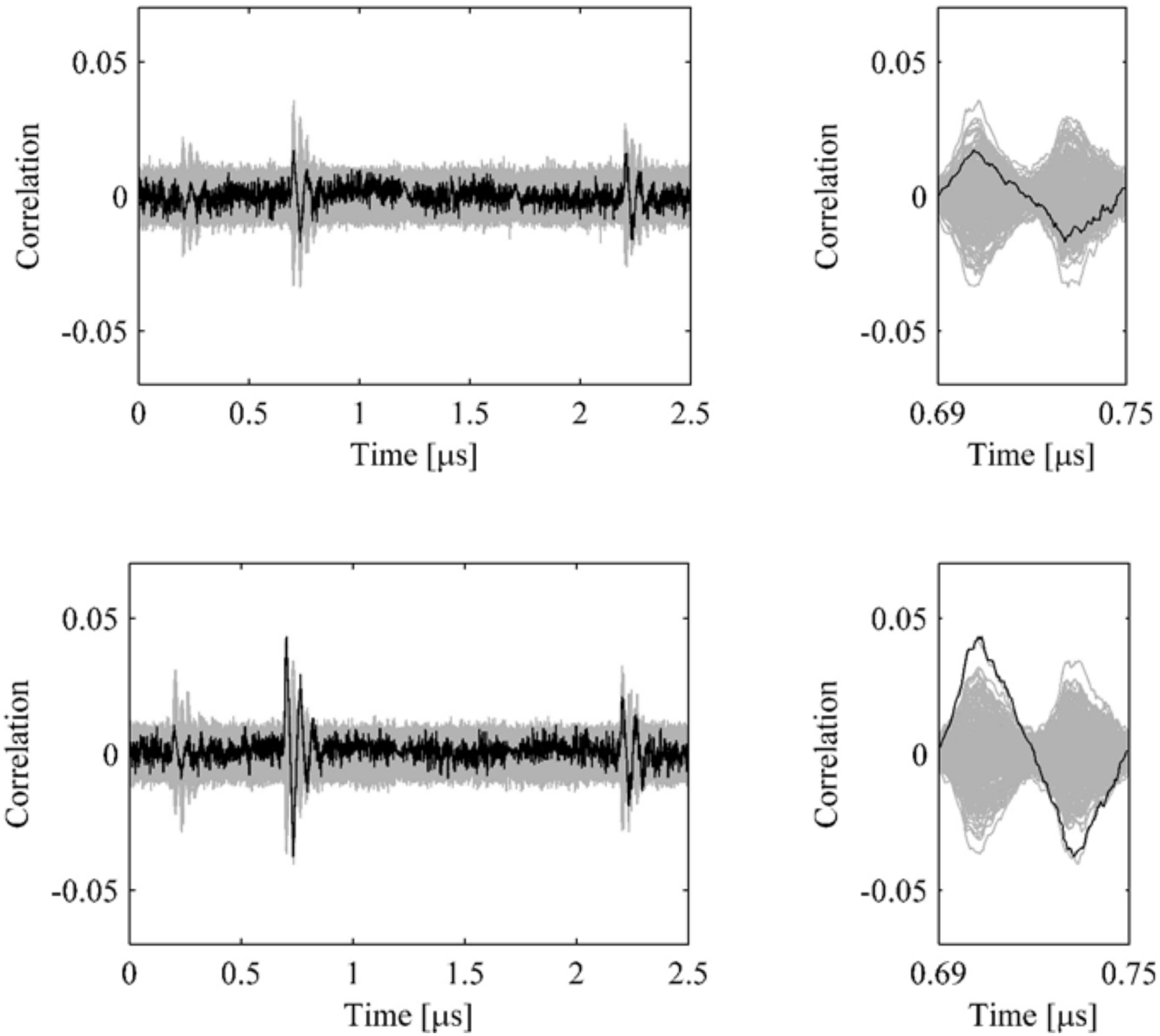
# Example of CPA on AES Hardware

- **Target intermediate value**
  - Inputs of the S-boxes in the last round

$$v_{i,j} = S^{-1}(d_i \oplus k_j)$$

- **100,000 random plaintexts**
- **Which power model to use?**

# Bit Model vs. HW Model




- **Bit model**
  - Fails to recover the key
- **HW model**
  - Succeeds with 50% prob.

# Hamming Distance Model

- Inputs of the S-boxes are stored in the same registers as their outputs

$$h_{i,j} = HD(v_{i,j}, d_i) = HW(v_{i,j} \oplus d_i)$$

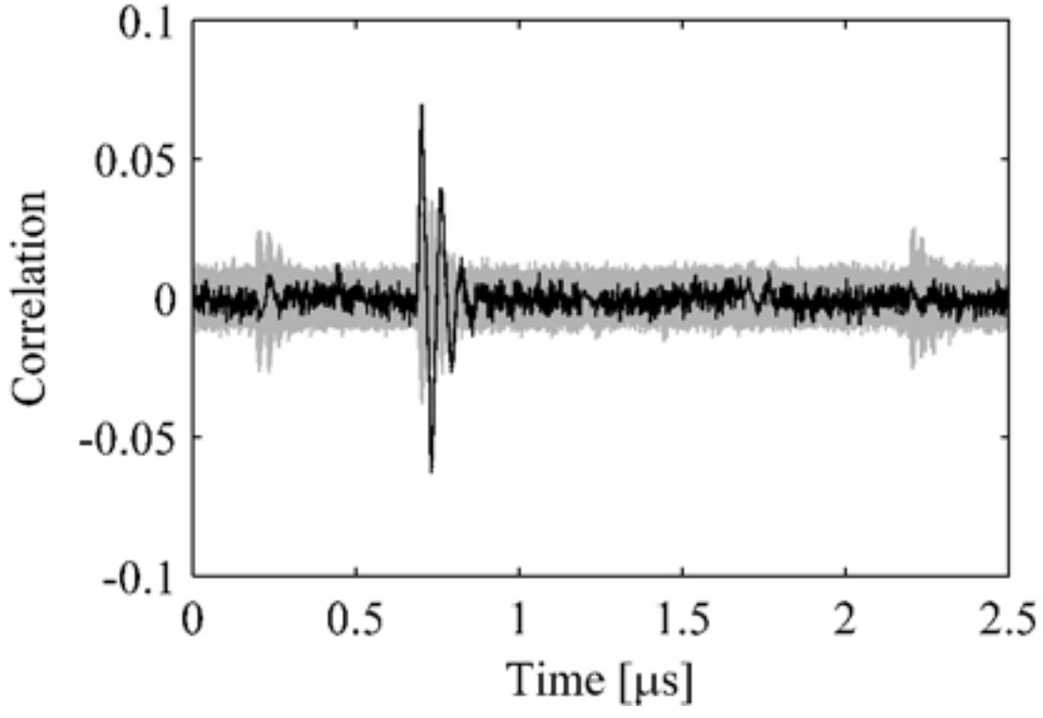



*Figure 6.12.* Result of the DPA attacks when using the HD model for byte one.

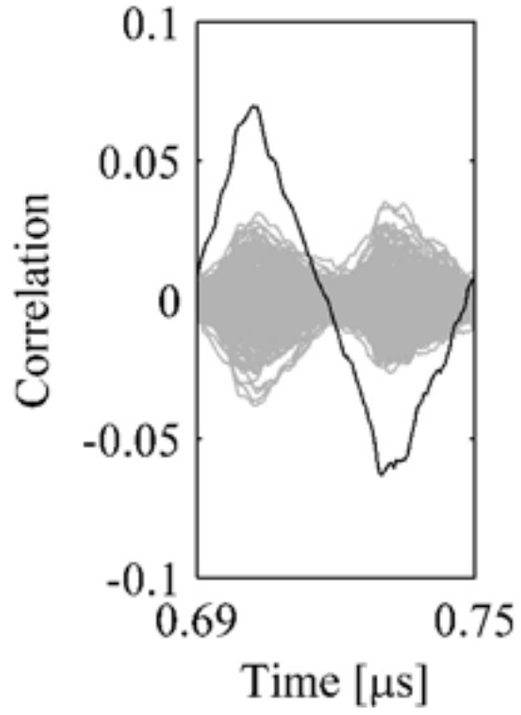
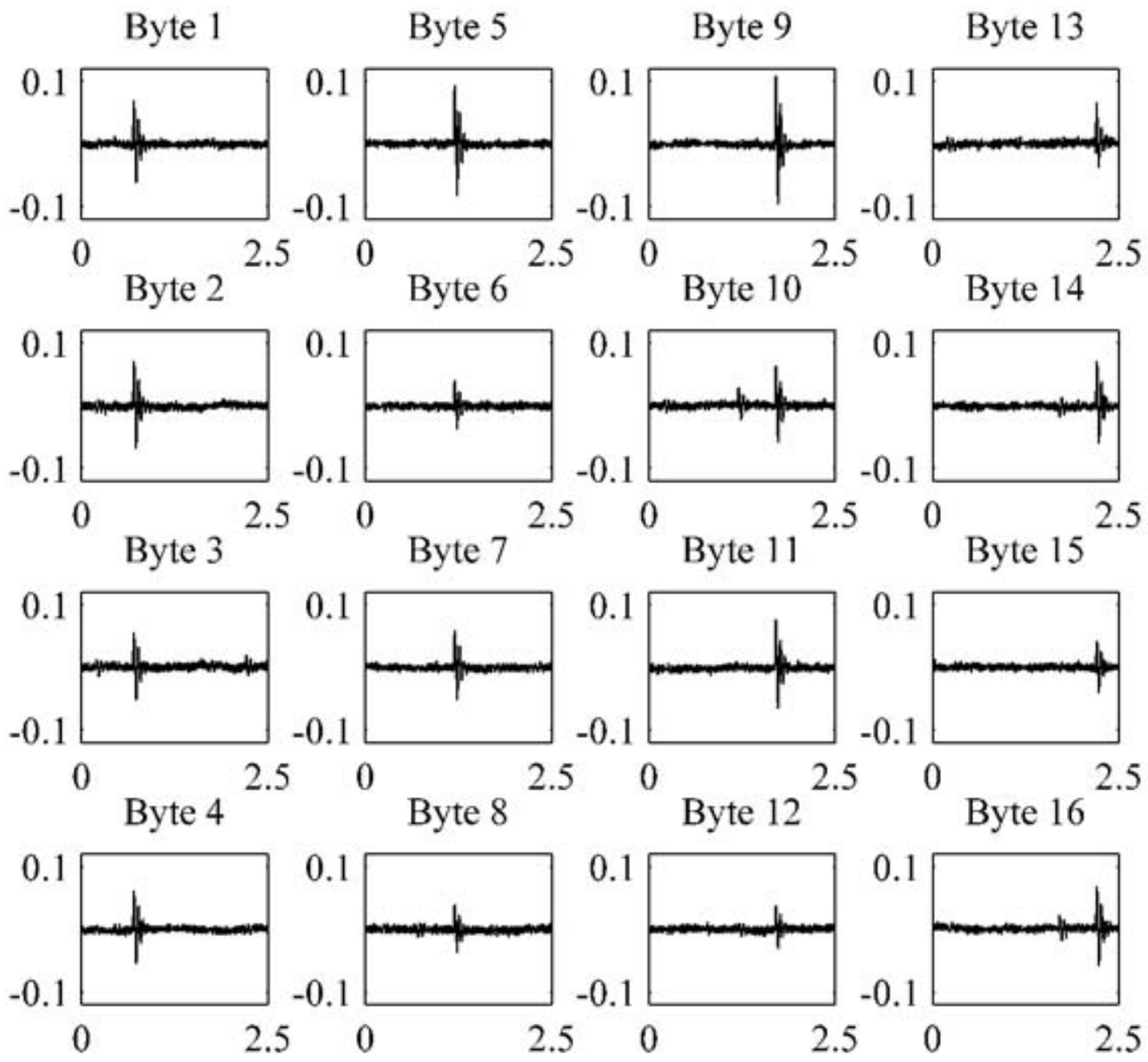



*Figure 6.13.* Zoomed view of the result.

# DPA Results on the 16 Key Bytes (Hardware)




- In each clock cycle, 4 S-box lookups are preformed in parallel.
- The correlation coefficients are strongly reduced because of the parallelism.
- The correlations of the 16 key bytes are significantly different.

*Figure 6.14.* Plots for the correct key hypotheses in the DPA attacks on the 16 S-box inputs in the last round. The y-axes show the correlation and the x-axes show the time in $\mu$s.

# Yet Another Power Model

- **Zero-value model**
  - In the AES hardware, the S-box are implemented based on composite field arithmetic.
  - Such implementation tends to consume significantly less power when the input is zero.

$$h_{i,j} = ZV(v_{i,j}) = \begin{cases} 0 & \text{for } v_{i,j} = 0 \\ 1 & \text{for } v_{i,j} \neq 0 \end{cases}$$

# DPA Using Zero-value Model

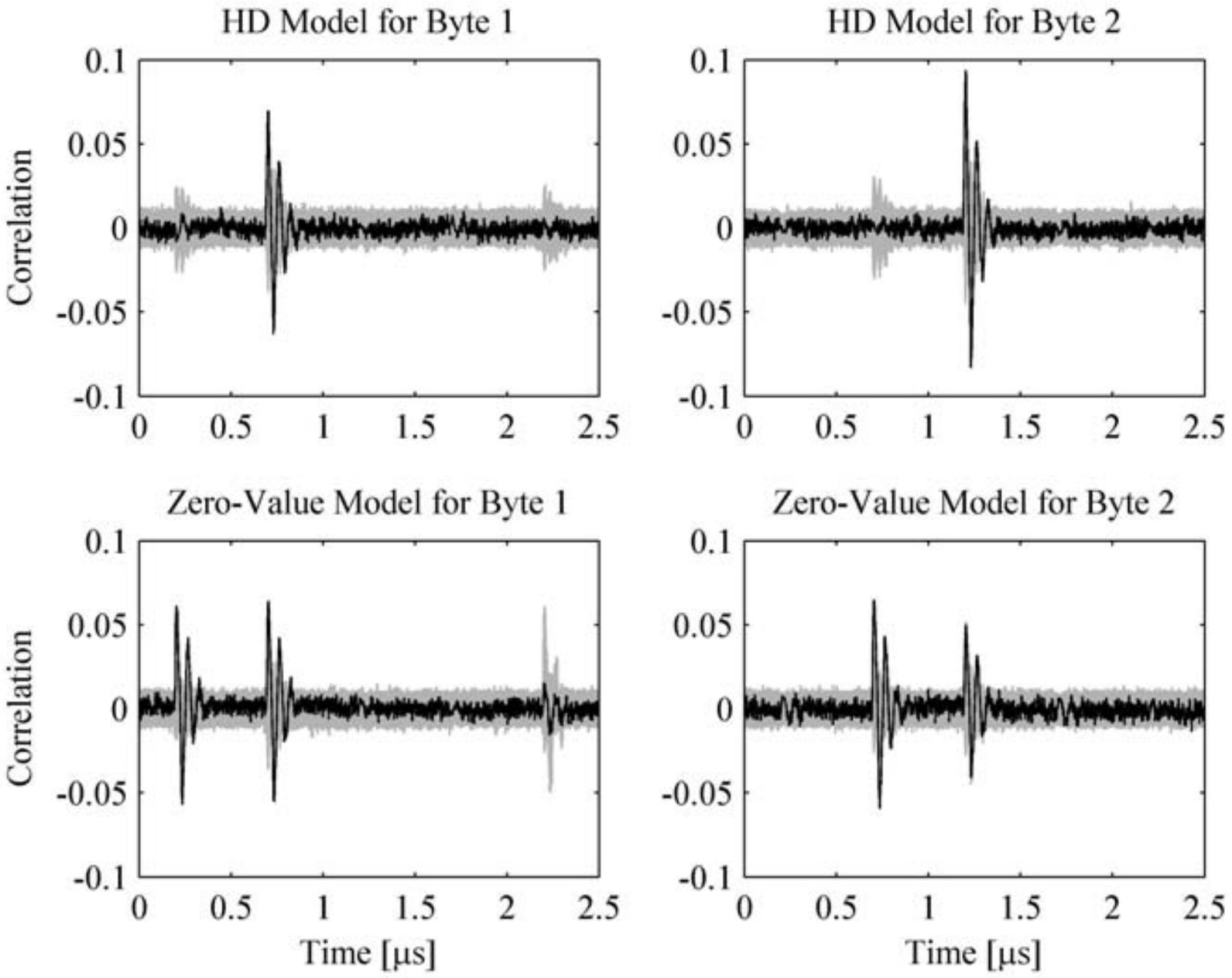



*Figure 6.15.* Results of the different DPA attacks on the input bytes of the first two S-boxes in the last round.

# Relationship between SNR and Correlation Coefficient

$$
\begin{aligned}
&\rho(H_i, P_{total}) = \\
&\quad = \rho(H_i, P_{exp} + P_{sw.noise} + P_{el.noise} + P_{const}) \\
&\quad = \rho(H_i, P_{exp} + P_{sw.noise} + P_{el.noise}) \\
&\quad = \rho(H_i, P_{exp} + P_{noise}) \\
&= \frac{E(H_i \cdot (P_{exp} + P_{noise})) - E(H_i) \cdot E(P_{exp} + P_{noise})}{\sqrt{Var(H_i) \cdot (Var(P_{exp}) + Var(P_{noise}))}} \\
&= \frac{E(H_i \cdot P_{exp} + H_i \cdot P_{noise}) - E(H_i) \cdot (E(P_{exp}) + E(P_{noise}))}{\sqrt{Var(H_i) \cdot Var(P_{exp})}\sqrt{1 + \frac{Var(P_{noise})}{Var(P_{exp})}}} \\
&= \frac{\rho(H_i, P_{exp})}{\sqrt{1 + \frac{1}{SNR}}}
\end{aligned}
\tag{6.5}
$$

# More on SNR and Correlation Coefficient

$$\rho(H_i, P_{total}) = \frac{\rho(H_i, P_{exp})}{\sqrt{1 + \frac{1}{SNR}}}$$

- The SNR is independent of the power model.
- How the leakage is exploited is described by $\rho(H_i, P_{exp})$
- In many cases, $\rho(H_i, P_{exp})$ can be determined by a simulated DPA.

# Example for Software

- **HW model**
  - The “simulated” power consumption

$$s_{i,ct} = HW(S(d_i \oplus k_{ck}))$$

  - In this case, it holds that

$$\rho(H_i, P_{exp}) = \rho(H_i, P_{data}) = \rho(H_i, \mathbf{s}_{ct})$$

  - Taking the noise into consideration, we have

$$\rho(H_i, \mathbf{t}_{ct}) = \frac{\rho(H_i, \mathbf{s}_{ct})}{\sqrt{1 + \frac{1}{22.89}}}$$

$$= \rho(H_i, \mathbf{s}_{ct}) \cdot 0.98$$

# Correlation Coefficients based on Simulated DPA

- This result shows the importance of the non-linear elements in DPA.

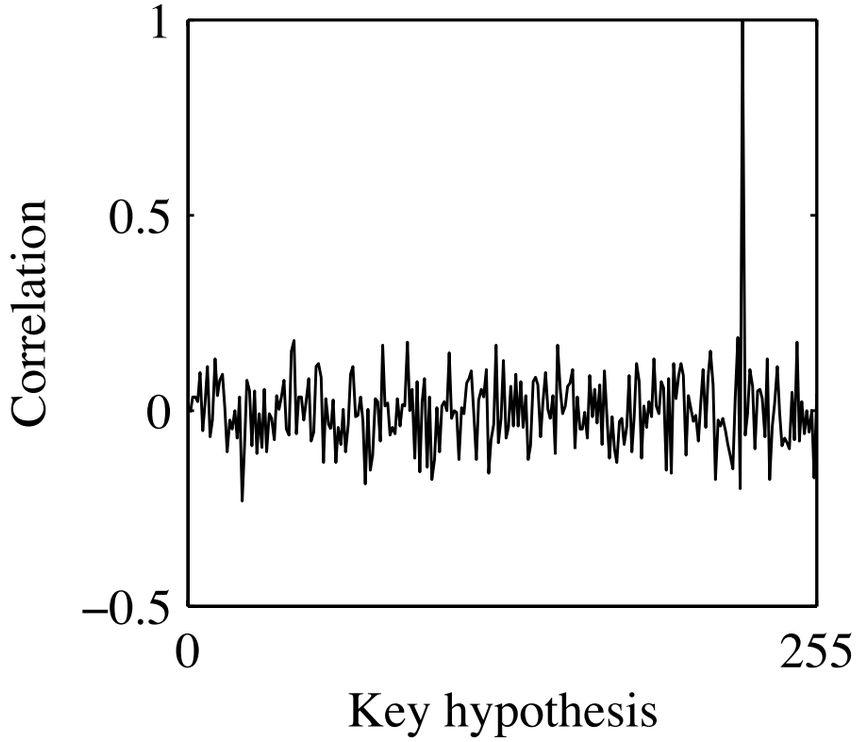



*Figure 6.16.* The correlation coefficients $\rho(H_i, P_{exp})$ occurring in a DPA attack using the HW model to attack an S-box output in the first round.

$$s_{i,ct} = HW(S(d_i \oplus k_{ck}))$$

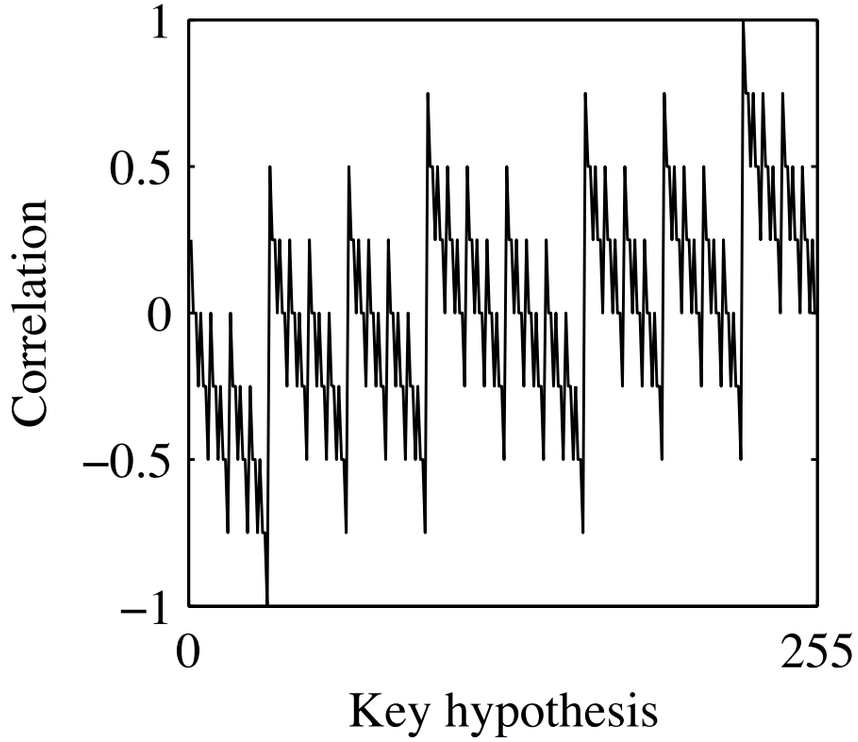



*Figure 6.17.* The correlation coefficients $\rho(H_i, P_{exp})$ occurring in a DPA attack using the HW model to attack an S-box input in the first round.

$$s_{i,ct} = HW((d_i \oplus k_{ck})$$

# How about the Bit Model?

- The simulated power $P_{exp}$ does not only depend on the attacked intermediate value, but also depends on the parts which is correlated to the attacked value.

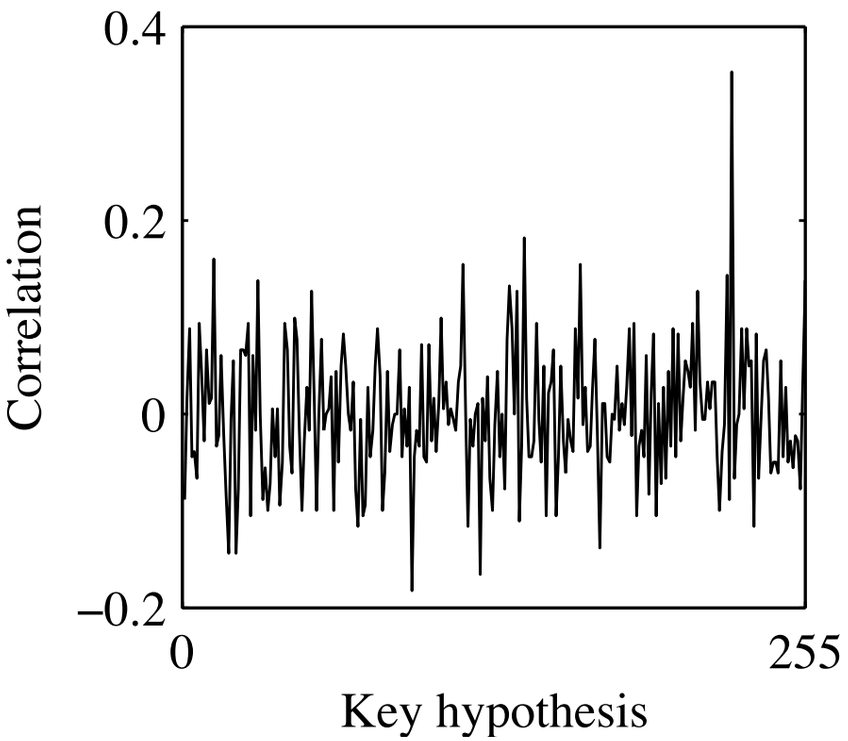



*Figure 6.18.* The correlation coefficients $\rho(H_i, P_{exp})$ occurring in a DPA attack using the bit model to attack the S-box output in the first round.

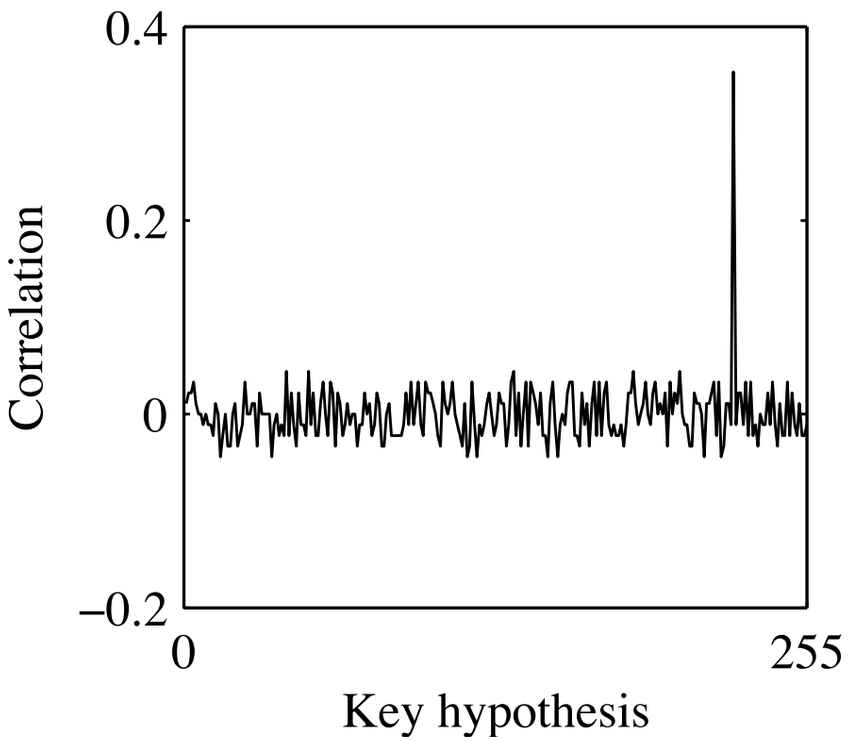



*Figure 6.19.* The correlation coefficients $\rho(H_i, P_{exp})$ when modeling the seven output bits of the S-box besides the LSB as switching noise.

Compute $\mathbf{s}_{i,ct}$ using the LSB only and treat the other 7 bits as noise. Therefore, we have

$$\rho(H_i, \mathbf{t}_{ct}) = \frac{\rho(H_i, \mathbf{s}_{ct})}{\sqrt{1 + \frac{1}{7}}} \cdot 0.98$$

# Example for Hardware

- **Much more complex simulation for hardware implementations**
  - Capacitive loads of the wires differ significantly
  - Data values processed in one clock cycle are partly correlated
    - Ex. AddRoundKey, SubByte, MixColumns are executed in one clock cycle on the AES hardware in this book.
  - The effect of glitches
- **No suitable power simulation for the attackers**

# Estimation Techniques for Attackers

- **Assumptions for simplification**
  - $\rho(H_i, P_{exp}) = 1$
  - All wires have the same capacitive load.
  - Switching activities of all wires are i.i.d.
- **A rough estimation**

$$\rho(H_{ck}, P_{total}) = \frac{\rho(H_{ck}, P_{exp})}{\sqrt{1 + \frac{1}{SNR}}} = \sqrt{\frac{\frac{a}{n}}{\frac{a}{n} + 1}} = \sqrt{\frac{a}{a+n}}$$

$a$: number of wires processing the attacked intermediate value.

$n$: number of wires processing statistically independent bits.

# Simulation Technique for Designers

- **Power simulation can be done very precisely by the designers, but can be very time consuming.**
  - We count the number of switchings instead.

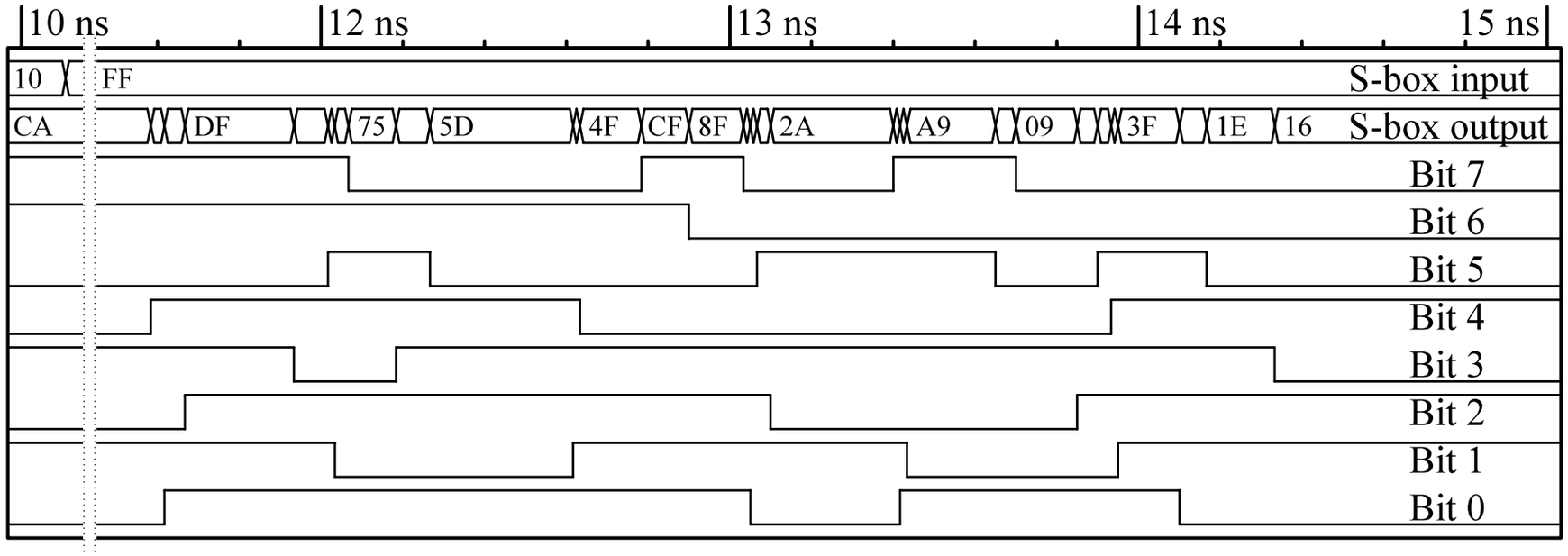



*Figure 6.20.* Simulation of the transitions occurring at the output of an AES S-box when the input changes from $10_{hex}$ to $FF_{hex}$.

# Simulated Results

- **Least transitions when the input is 0.**
  - This explains why the zero-value model works.

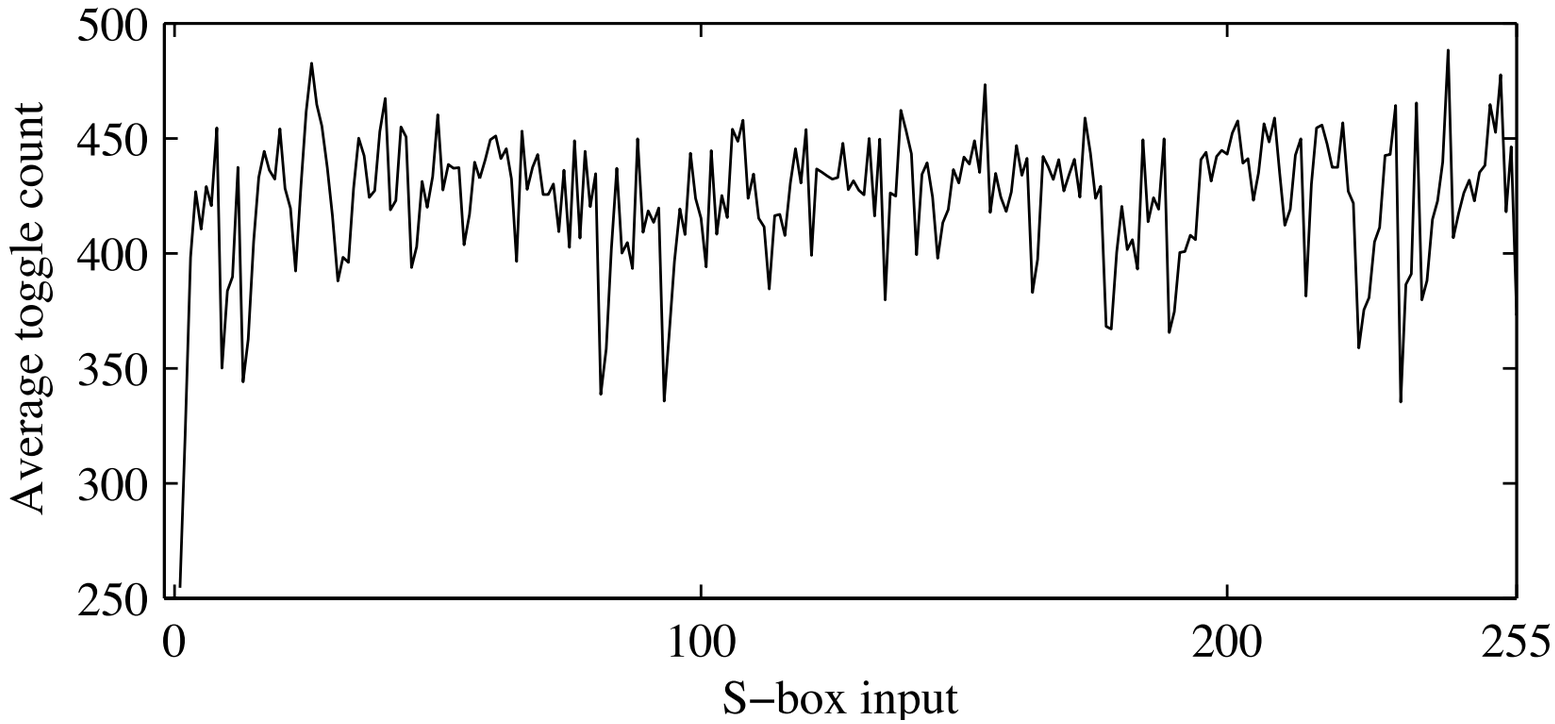



*Figure 6.21.* Average number of transitions occurring in the S-box for the 256 input values.

# Explanation for the Failure of Bit & HW Model

- The differences between the peak and the second highest value are very small.
  - 0.018 vs. 0.025

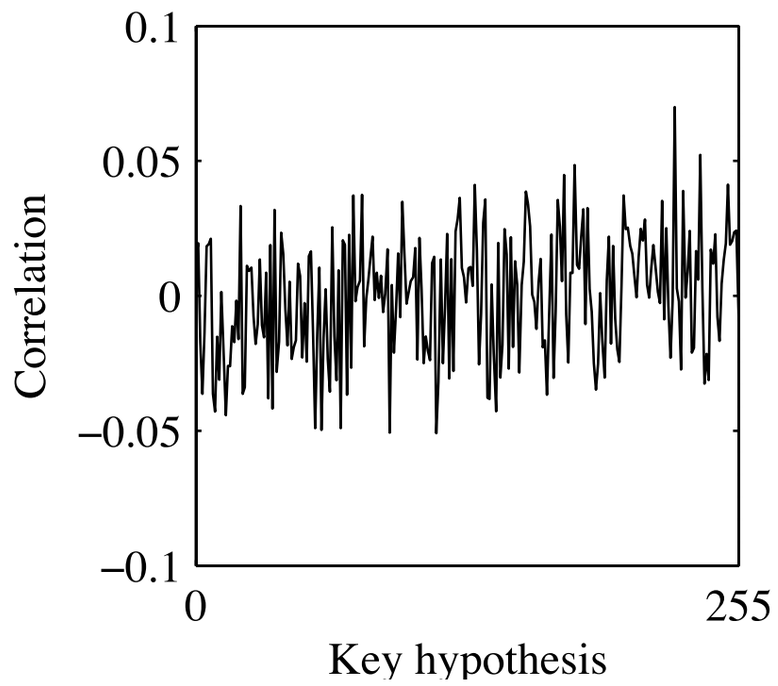



*Figure 6.22.* Correlation coefficients that occur when using the HW model to perform a DPA attack on the simulated toggle counts of the S-box.

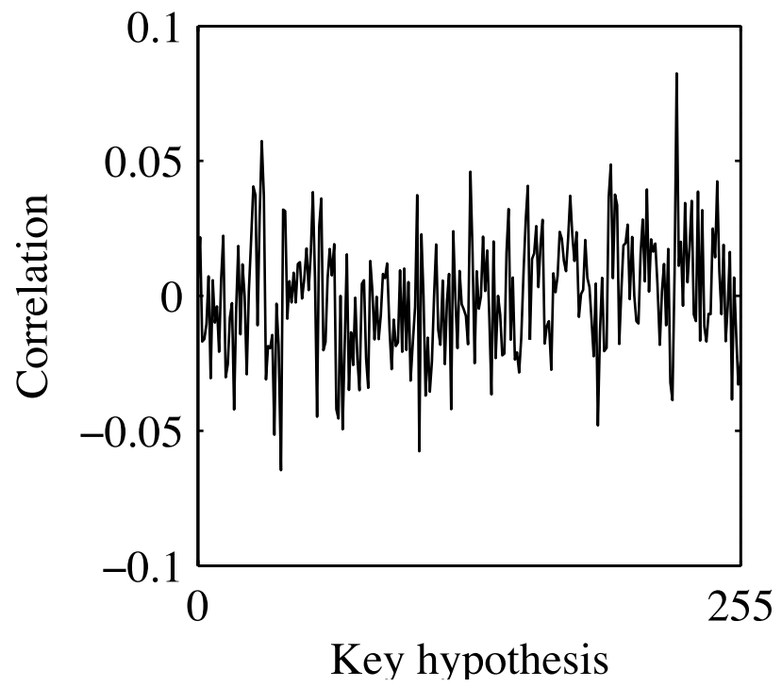



*Figure 6.23.* Correlation coefficients that occur when using the bit model to perform a DPA attack on the simulated toggle counts of the S-box.

# How many power traces do we need?

- A DPA is successful if there is a significant peak in the correlation coefficients when the key is correct.
- The number of traces for distinguishing the estimated correlation coefficient $\rho_{ck,ct}$ from an estimated $\rho = 0$

# Let's recall some statistics again

- Fisher's transformation of the correlation coefficient

$$Z_1 = \frac{1}{2} \ln \frac{1+R}{1-R}$$

$$\mu = \frac{1}{2} \ln \frac{1+\rho}{1-\rho} + \frac{\rho}{2(n-1)}$$

$$\sigma^2 = \frac{1}{n-3}$$

$$\Rightarrow Z = \frac{Z_1 - \mu}{\sigma} \sim \mathcal{N}(0,1)$$

- The number of traces to distinguish an estimator of a standard normal distribution from zero can be derived from

$$\Pr\{|\bar{X} - \mu| > c\} = \Pr\{|Z| > \frac{c\sqrt{n}}{\sigma}\} = \alpha$$

# Rule of Thumb

$$n = 3 + 8\frac{z_{1-\alpha}^2}{\ln^2 \frac{1+\rho_{ck,ct}}{1-\rho_{ck,ct}}}$$

| $\rho_{ck,ct}$ | $n_{\alpha=0.0001}$ | $\rho_{ck,ct}$ | $n_{\alpha=0.0001}$ | $\rho_{ck,ct}$ | $n_{\alpha=0.0001}$ |
|---|---|---|---|---|---|
| 0.900 | 16 | 0.090 | 3 400 | 0.009 | 341 493 |
| 0.800 | 26 | 0.080 | 4 307 | 0.008 | 432 206 |
| 0.700 | 40 | 0.070 | 5 630 | 0.007 | 564 519 |
| 0.600 | 61 | 0.060 | 7 668 | 0.006 | 768 378 |
| 0.500 | 95 | 0.050 | 11 049 | 0.005 | 1 106 471 |
| 0.400 | 157 | 0.040 | 17 273 | 0.004 | 1 728 870 |
| 0.300 | 292 | 0.030 | 30 720 | 0.003 | 3 073 559 |
| 0.200 | 676 | 0.020 | 69 140 | 0.002 | 6 915 526 |
| 0.100 | 2 751 | 0.010 | 276 606 | 0.001 | 27 662 152 |

For $|\rho_{ck,ct} \leq 0.2|$ and for small SNRs, the following relations hold:

$$n \approx \frac{28}{\rho_{ck,ct}^2} \sim \frac{1}{SNR} = \frac{Var(P_{sw.noise} + P_{el.noise})}{Var(P_{exp})} \tag{6.9}$$

# Example

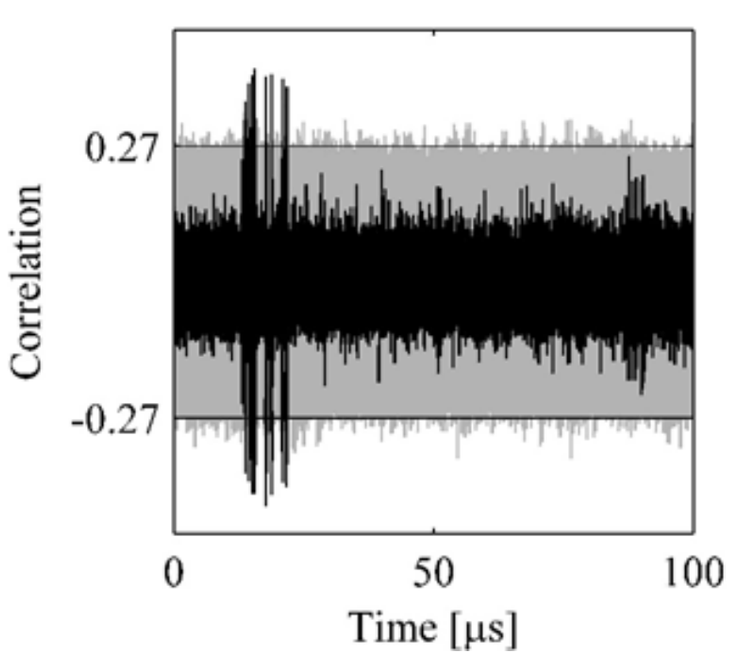



*Figure 6.24.* Result of the DPA attack on the microcontroller when using 224 traces.

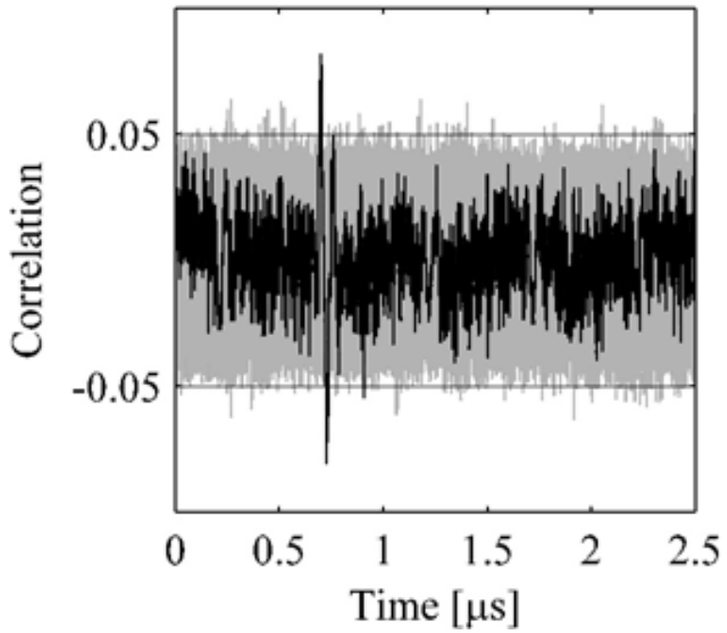



*Figure 6.25.* Result of the DPA attack on the AES ASIC implementation when using 6 532 traces.

$$\rho_{ck,ct} = 0.34$$
$$\Rightarrow n = 224$$
$$\Rightarrow \frac{4}{\sqrt{n}} = 0.27$$

$$\rho_{ck,ct} = 0.065$$
$$\Rightarrow n = 6532$$
$$\Rightarrow \frac{4}{\sqrt{n}} = 0.05$$

# Alternatives to the Correlation Coefficient

- **Difference of means**

$$
\begin{aligned}
m_{1i,j} &= \frac{1}{n_{1i}} \cdot \sum_{l=1}^{n} h_{l,i} \cdot t_{l,j} \\
m_{0i,j} &= \frac{1}{n_{0i}} \cdot \sum_{l=1}^{n} (1 - h_{l,i}) \cdot t_{l,j} \\
n_{1i} &= \sum_{l=1}^{n} h_{l,i} \\
n_{0i} &= \sum_{l=1}^{n} (1 - h_{l,i}) \\
\mathbf{R} &= \mathbf{M_1} - \mathbf{M_0}
\end{aligned}
$$

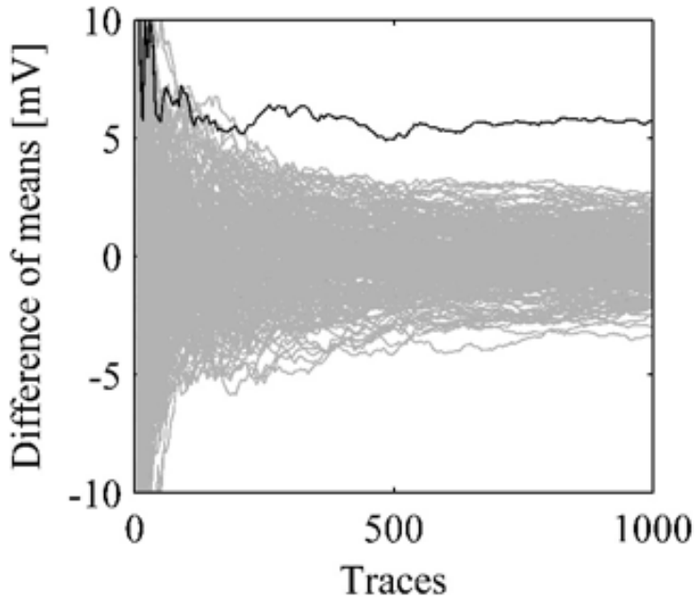
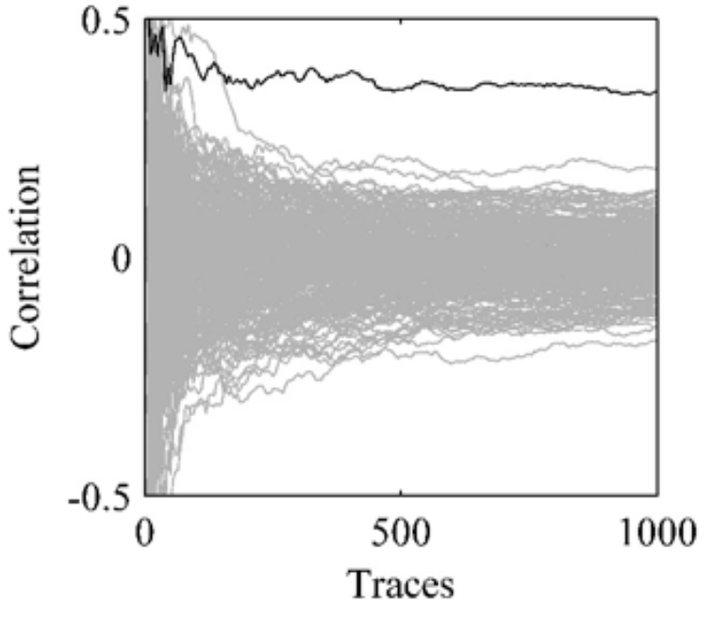



*Figure 6.27.* The column of $\mathbf{R}$ at 13.8 $\mu$s for different numbers of traces. Key hypothesis 225 is plotted in black.



# Alternatives to the Correlation Coefficient

- **Distance of means**
  - Difference of means only considers the mean values.
  - Take the variances into account

$$r_{i,j} = \frac{m_{1i,j} - m_{0i,j}}{s_{i,j}}$$

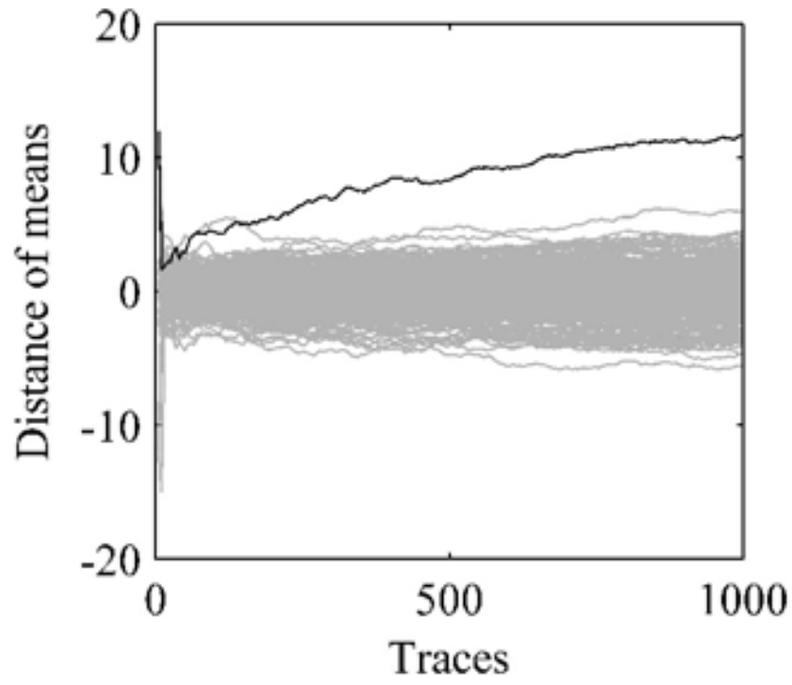



*Figure 6.29.* The column of $\mathbf{R}$ at 13.8 $\mu$s for different numbers of traces. Key hypothesis 225 is plotted in black.